



Research article



Blockchain-enabled traceability for rice blending fraud: A practical framework to strengthen supply chain integrity in smart agricultural IoT

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ABSTRACT

The rising demand for high-quality rice, driven by improving living standards, has led consumers to increasingly pay premium prices. However, the scarcity of high-quality rice resources has created opportunities for adulteration, often motivated by profit-seeking behavior. Blockchain technology, characterized by its tamper-proof nature, transparency, and traceability, presents a robust solution to this challenge. This paper proposes a blockchain-based framework to address the issue of mixed selling of rice of varying qualities, thereby enhancing supply chain efficiency and credibility. The proposed solution is designed to be generalizable, applicable to products with similar forms yet significantly different qualities. Empirical results show that the data throughput of the proposed model scales linearly with increasing test size, highlighting its scalability and stable performance in large-scale implementation. This study underscores the potential of blockchain technology to mitigate adulteration risks while supporting scalable and reliable supply chain management.

1. Introduction

Recent security incidents have drawn public attention [1,2]. Rice is a globally consumed staple, and its quality and safety impact consumer health. The uniform appearance and large volume of rice grains make visual quality distinction difficult, resulting in market mixing of different quality rice. Poor-quality rice can cause health issues. Reports indicate that fraud in the rice supply chain has raised concerns. Farms may produce sub-par high-quality rice or even alter their data [3]. Unqualified processing companies might mix rice of varying qualities during packaging [4]. These challenges are further exacerbated by the globalization of food production, which introduces additional complexities into supply chains and makes ensuring food quality and traceability increasingly difficult [5]. Currently, traditional information management systems are still widely used in agricultural production and processing, with manual recording in some cases [6]. Within the rice supply chain, each entity often uses independent databases, leading to severe information silos [7]. The fragmented and incompatible data storage methods exacerbate the lack of trust mechanisms and data-sharing difficulties in the supply chain.

Technological advancements, including blockchain, real-time monitoring, and augmented reality, are emerging as critical drivers for enhancing food safety and restoring consumer trust [8,9]. Blockchain technology, in particular, establishes a tamper-resistant and transparent system for all stakeholders through its immutable data-sharing mechanisms and distributed ledger infrastructure. This fundamentally ensures the permanent traceability of transaction records and universal data accessibility [10–13]. The viability of

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this technological paradigm is demonstrated by its successful application in high-stakes business and legal process registration [14], and its provision of a reliable framework for traceability and transactions in sustainable energy systems and the circular economy, such as in electric vehicle battery recycling and smart grids [15].

Within tangible industrial sectors, the deep integration of blockchain with technologies like the Internet of Things (IoT) and Artificial Intelligence (AI) is propelling fields such as construction safety and food traceability toward data-driven, automated management [16,17]. Furthermore, the incorporation of AI augments these systems with intelligent decision-making capabilities, enabling real-time quality assessment, automated abnormal product recalls, and fostering new models for standardized regulation through its synergy with blockchain audit logs [18,19].

However, the widespread adoption of blockchain is not solely a technical endeavor; its success is critically influenced by non-technical factors. Existing research clearly identifies barriers such as data privacy, scalability, and the absence of legal frameworks as primary challenges [20], while economic viability and potential market-structural conflicts represent significant variables affecting its implementation [21,22]. The collective insights and successful practices from these diverse fields provide a solid theoretical foundation for this study's exploration of blockchain in the agricultural product traceability domain, while also illuminating critical implementation pathways and risk mitigation strategies.

This study advances supply chain management by resolving critical quality and traceability challenges through a novel Fisco Bcos blockchain model. The framework integrates consumer feedback to enhance supply chain sustainability, with empirical validation confirming its real-world efficacy. Additionally, it proposes a generalized blockchain solution adaptable to diverse products (e.g., wine, electronics, and luxury goods), overcoming traditional limitations like static tracking, data silos, and tampering risks. By improving transparency and operational trust, the model demonstrates broad industrial applicability. Furthermore, it provides a robust framework for ensuring food safety and quality in the food supply chain, thereby making a significant contribution to food engineering.

The remainder of this paper is organized as follows: Section 2 discusses the current situation of blockchain applications and rapid detection technology for rice. Section 3 demonstrates the designs of the proposed scheme. Section 4 presents the results and discussions. Finally, the conclusions are drawn in Section 5.

2. Related works

In recent years, blockchain technology has attracted considerable attention due to its significant potential for application across complex, multi-domain systems, positioning it as a key enabler for future technological ecosystems [23]. This work aims to provide a high-performance blockchain solution for crop supply chain management scenarios. In order to ensure the quality requirements of rice, several quality tests are required during this research. Therefore, we will focus on two key aspects: 1) the application of blockchain in supply chain management; and 2) the implementation of rapid inspection techniques in the rice supply chain.

2.1. Blockchain for supply chain management

Zhang et al. [24] advocated leveraging blockchain technology to modernize the fresh fruit supply chain, underscoring its benefits in management. Blockchain, in tandem with IoT, can gather and upload trustworthy data from fruit harvesting to consumption, with hash encryption and consensus mechanisms ensuring data security. The research indicates that blockchain balances supply and demand, provides financial backing to businesses, simplifies regulatory procedures, and enhances supply chain efficiency by reducing communication and trust costs via smart contracts.

Rehman et al. [25] presented a smart agriculture model utilizing blockchain technology and IoT with the aim of improving the accuracy and credibility of data management. The results of the study showed that by installing sensors to monitor temperature, humidity, and light, the use of agrochemicals can be controlled, and timely alert notifications can be issued. The proposed protocol contributes to energy conservation and enhances the sustainability of agricultural operations. Meanwhile, the application of blockchain technology in agriculture is expected to enable smart agricultural contracts and streamline management processes. The research results show that the model is important for the decision-making process, promoting water-efficient crop cultivation, weather adaptation and sustainability, which in turn is expected to address the current food safety challenges.

Yu et al. [2] systematically reviewed blockchain-enabled traceability systems for fresh food quality, emphasizing the integration of spectroscopy, mass spectrometry, and DNA-based biomarkers to generate immutable product fingerprints. By embedding sensor-derived environmental parameters (e.g., temperature, humidity, and gas concentrations) and biochemical signatures into distributed ledgers, the framework establishes multi-dimensional verification mechanisms across production, logistics, and retail phases. Dynamic monitoring via intelligent films and electronic noses enhances real-time quality assessment, while RFID-NFC-QR code triplex labeling enables granular traceability from farm to consumer. The study highlights blockchain's capacity to mitigate adulteration risks through cryptographic anchoring of geographical origin data, particularly in cereal crops like rice, where spectral profiles and isotopic ratios serve as discriminative markers. However, scalability constraints persist due to heterogeneous data formats and computational overhead in large-scale operations. These findings provide actionable insights for combating grain commingling by establishing decentralized authentication protocols, where blockchain's tamper-evident records, coupled with chemometric verification, could enforce strict origin segregation in bulk rice supply chains.

Pang et al. [26] conducted a systematic evaluation of blockchain-based agricultural traceability systems, emphasizing the technical challenges posed by bulk product blending and heterogeneous data formats. Their framework integrates product identification

modeling with blockchain-IoT-AI tripartite verification to address traceability disconnections, demonstrating that permissioned consortium chains achieve stable throughput under a certain workload. The study identifies standardization gaps in multi-stakeholder data sharing protocols, particularly in export-oriented supply chains, and proposes granular compliance digitization through smart contract automation. Research on typical supply chains reveals that a majority of implementation barriers stem from inconsistent ERP interoperability and fragmented regulatory frameworks. These findings underscore the necessity of government-led traceability alliances to harmonize blockchain architectures with existing quality certification systems, thereby enabling batch-level origin segregation critical for combating grain commingling.

Ferrández-Pastor et al. [27] proposed an agricultural traceability model based on user-centered design, value chain planning and digital technologies based on the IoT and blockchain technologies. In this, the user-centered design approach involves the participation of agricultural experts to determine the adaptation and selection of digital services. IoT technology is used to capture and process physical variables in the agricultural process and provide a network for data sensorization, control and storage. Blockchain technology is then used to provide tamper-proof, transparent and secure traceability services to ensure quality and safety controls in the value chain. The combination of these approaches aims to provide a comprehensive digital solution for the agricultural sector to optimize agricultural processes and provide traceability services.

Hajji et al. [28] developed a blockchain-based management system for agrifood supply chains using the Hyperledger Fabric framework. By integrating chaincode logic and membership service providers, the system enforces multi-tiered access control to ensure traceability and operational compliance. The architecture employs a builder pattern for dynamic generation of commodity batch units and incorporates one-time agreement mechanisms to mitigate replay attack risks. Real-time logistics tracking is enabled through a REST API and a web platform, allowing stakeholders to monitor product movements via QR code scanning. Performance evaluations under a certain workload demonstrated stable throughput with acceptably low latency, validating the feasibility of consortium blockchain in complex supply chain scenarios. This framework addresses data fragmentation in agricultural distribution and offers methodological insights for traceability governance in rice supply chains, particularly in combating adulteration through authenticated, end-to-end accountability mechanisms.

Current research on blockchain in the food industry primarily centers on food traceability, technology upgrade, and consumer acceptance [29,30]. Our research innovatively integrates consumer decision participation with multi-stage verification in circulation. This strengthens mutual accountability, encourages due diligence, and curbs fraud. Consumer involvement also heightens supply chain transparency and trust. Overall, this approach overcomes existing limitations and offers a more efficient and reliable agricultural supply chain management model.

2.2. Rapid detection technologies in rice supply chain management

In the rice supply chain, ensuring product quality through rapid and efficient quality detection is of paramount importance. Emerging technologies such as near-infrared spectroscopy (NIR), Raman spectroscopy, and molecular markers provide innovative solutions for the rapid detection of agricultural product quality. Pasquini et al. [31] employed compact NIR spectroscopy combined with chemometrics to achieve non-destructive detection of the swell volume and identification of adulteration in psyllium husk, revealing the market phenomenon where price does not correspond to quality. In contrast, Wang et al. [32] significantly enhanced the generalization capability and accuracy of a miniature NIR spectrometer in predicting key component contents in various tea types by adopting a fine-tuning transfer strategy based on deep learning.

Aznan et al. [33] introduced a novel approach for detecting rice adulteration using low-cost, portable devices such as a handheld NIR spectrometer and e-nose sensors, coupled with machine learning algorithms. This method demonstrated high accuracy and efficiency in identifying adulterated rice samples through packaging, offering a practical and cost-effective solution for rice quality monitoring. These advancements collectively represent significant progress in the application of rapid detection technologies in the rice supply chain, enhancing both efficiency and reliability.

Wang et al. [34] demonstrated the feasibility of non-destructive internal crack detection in rice seeds using NIR spectroscopy and chemometrics. Among the evaluated algorithms, partial least squares discriminant analysis applied to raw spectra achieved superior performance. The study identified key wavelengths associated with amylose content as critical for discrimination. This approach presents a rapid and reliable method for screening seed internal defects, showing significant potential for enhancing quality control in the seed industry.

3. Proposed scheme

For agricultural products, this study proposes a blockchain-based supply chain management model. The goal of the supply chain architecture is to accurately track rice throughout the chain and ensure consistency and stability of its quality.

3.1. Rice circulation process

In our proposed model, the rice traceability process is segmented into key stages: production, processing, transportation, retailing, and government regulation. The distribution process of rice is illustrated through a business process model. Fig. 1 outlines the blockchain-based supply chain process model for rice blending fraud prevention, encompassing multiple entities such as regulators, farms, factories, distributors, retailers, and final consumers, clearly reflecting the entire journey of rice from production to consumption.

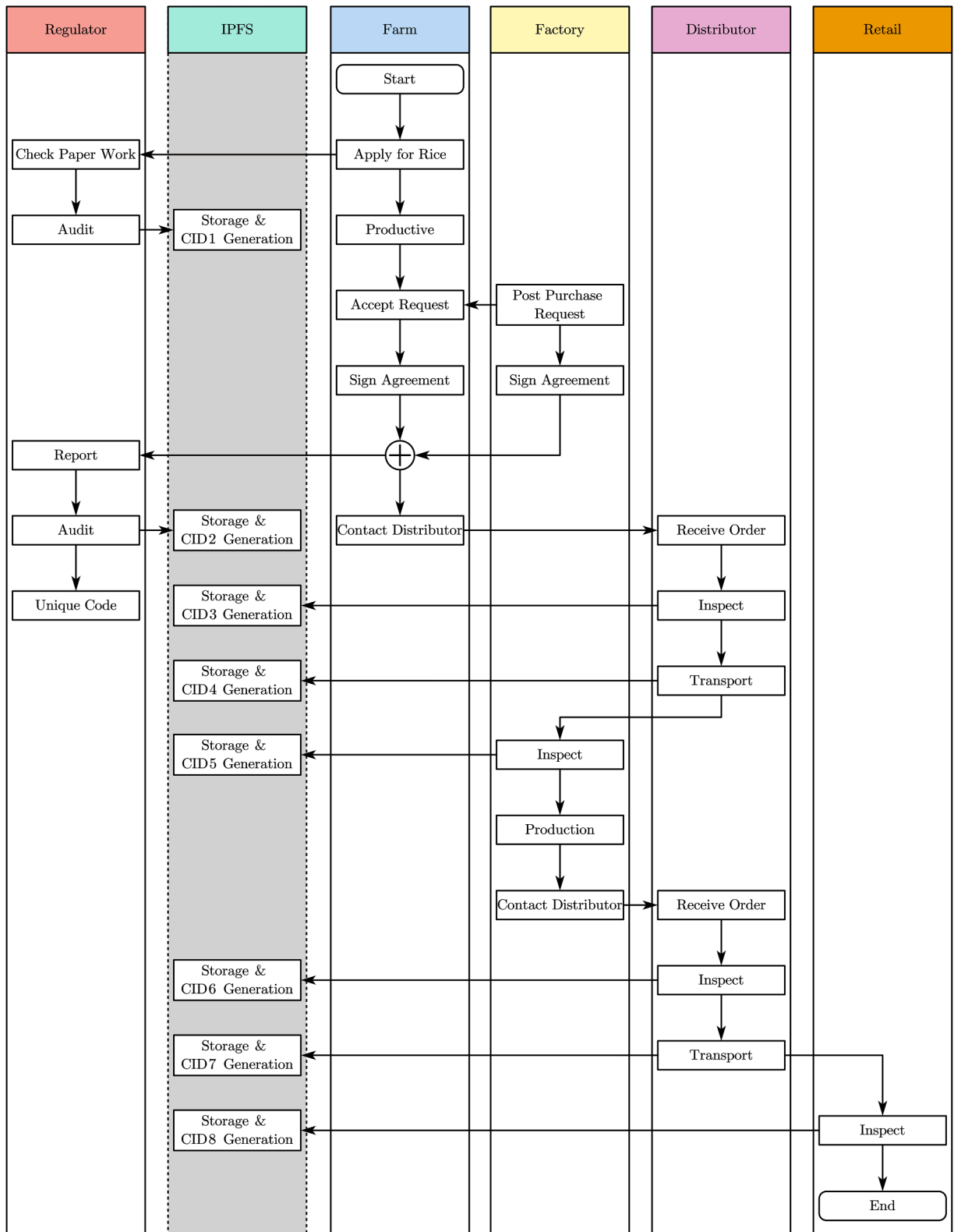


Fig. 1. Blockchain-based supply chain process model for rice blending fraud prevention

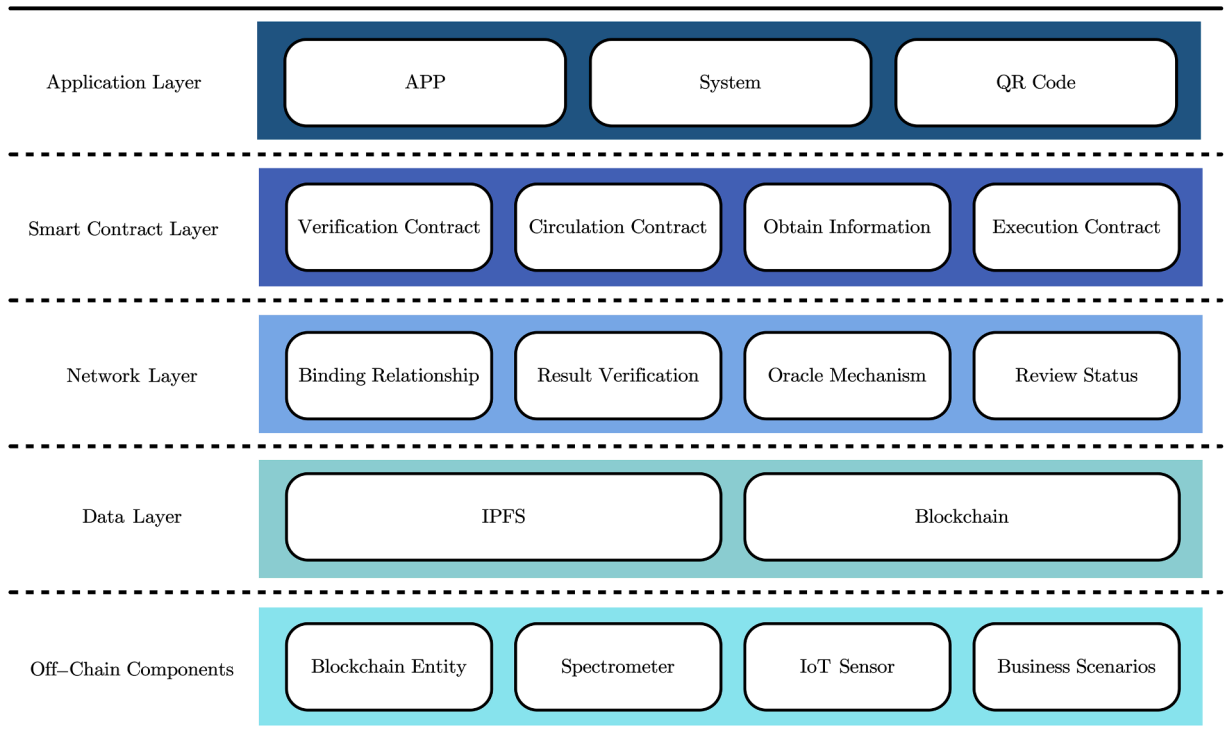


Fig. 2. Blockchain-based hierarchical system architecture design for rice blending fraud prevention

Before circulation, all entities must apply for a regulatory filing with the regulator. Subsequent supply chain processes are permitted only after the regulator has vetted and approved these entities. The regulator, typically a government agency, is responsible for verifying entity qualifications and inspecting rice quality.

Once farm production is completed, farmers submit rice samples and planting records to the regulator. The regulator uses an NIR spectrometer to assess rice quality, analyzing elements like starch content and heavy metal residues. This process generates spectral data, including wavelength-absorbance curves, which the regulator leverages to rate the rice (high, medium, or low). Following regulatory approval, the spectral data and farm information are uploaded as separate datasets to the InterPlanetary File System (IPFS), each generating a unique content identifier (CID). All CIDs are recorded on the blockchain, with the spectral data identifier specifically designated as CID1 for reference purposes.

Factories access rice-related information on the blockchain and reach out to farms to sign purchase contracts. These contracts are then submitted to the regulator for review. Once approved, the regulator uploads the contract to IPFS, generating CID2, and assigns a unique number that is linked to CID1.

Once the distributor accepts the rice, it uses a NIR spectrometer for testing, generating CID3. The submission of CID3 automatically triggers a smart contract to compare it with the benchmark CID1. If verified, the distribution process commences. During transportation, IoT sensors installed on the transport vehicle collect data in real-time.

Upon the arrival of the shipment, sensor data from the transport vehicle is automatically uploaded to IPFS, generating CID4. Subsequently, the factory conducts an NIR spectrometer test on the rice and uploads the results to IPFS, generating CID5. The submission of CID5 triggers a smart contract, which automatically compares CID5 against the benchmark CID1 for quality verification. Only after successful validation is the facility authorized to proceed with the next production steps.

After processing, the factory notifies the distributor. The distributor's NIR test results for the processed rice are submitted for verification, with the upload to IPFS generating CID6. This automatically triggers the smart contract to compare CID6 with CID1, authorizing shipment to the retailer upon a successful match.

Upon arrival, vehicle sensor data is uploaded to IPFS, generating CID7. The retailer then conducts an NIR test and uploads the results to generate CID8. The submission of CID8 automatically triggers a smart contract to compare it with the benchmark CID1. A successful match confirms product quality, completing the supply chain and authorizing the rice for sale.

After purchasing rice, consumers can scan a QR code to access comprehensive traceability information. Should they have any quality concerns, they can submit an inquiry to the regulator, supported by photographic evidence and descriptions. Upon receiving a complaint, the regulator will inspect the specified batch using an NIR spectrometer. The results are compared against the benchmark

CID1, and the investigation is permanently recorded on the blockchain. If validated, the consumer is rewarded and the responsible retailer is penalized.

3.2. Model architecture diagram

Fig.2 illustrates blockchain-based hierarchical system architecture design for rice blending fraud prevention. This design encompasses multiple layers and components. Each layer and component play a vital role in ensuring the model's functionality, reliability, and effectiveness in addressing the challenges of rice supply chain traceability.

The Application Layer supplies user-friendly interfaces for all participants, enabling data upload, inquiry, and transactions. Farms, factories, distributors, and retailers can upload data, spectral information, and check the real-time status. Consumers scan QR codes to access full-traceability rice data which can raise quality concerns. Regulators use the model to audit farm data, rate rice quality, and address consumer complaints. This layer ensures transparency and efficiency across the supply chain.

Functioning as the model's core logic engine, the Smart Contract Layer performs spectral data validation and sensor recording to verify product integrity. It governs the rice circulation process, provides real-time status and entity information queries, and enforces regulatory compliance by suspending non-conforming batches and executing penalties following inspections.

The Network Layer overcomes blockchain's inherent limitation in accessing off-chain data by implementing an oracle mechanism to retrieve and computationally validate spectral data stored in IPFS. This layer also manages the association of spectral documents and organizes information throughout the circulation process.

The Data Layer employs a hybrid architecture integrating IPFS and Blockchain to fulfil distinct data management requirements. IPFS is tasked with storing unstructured data, such as spectral datasets, procurement contracts, and sensor-generated reports, with each asset being assigned a unique CID. The Blockchain complements this by serving as a secure ledger for the CIDs, critical traceability events, and state transitions, all managed via smart contracts. This synergistic design not only enhances auditability and trust across the rice supply chain but also improves storage efficiency and system scalability.

The Off-Chain Components bridge the model with the physical world, encompassing NIR spectrometers for rice quality testing, IoT sensors for real-time environmental monitoring on transport vehicles. It has potential for future expansion into commercial services such as gas stations, repair shops, and insurance companies.

3.3. Model design

Fig.3 depicts the blockchain-based underlying data model for rice blending fraud prevention and traceability, illustrating the key entities' attributes, behaviors, and their interrelationships. Each participating entity has a unique name and participates by invoking functions in the smart contract. Fig. 4 outlines the blockchain-based sequence flow of rice circulation for blending fraud prevention. Since the farmer and the factory have signed a purchase contract prior to this, the farmer needs to contact the distributor to transport the rice. The farmer calls the `getTransportList()` method on the blockchain network, which returns all the vetted distributors and their details. Based on the returned results, the farmer can choose and contact the appropriate distributor. In this supply chain process, the distributor executes the function `insertByTransport()`, which includes the following key attributes: rice purchase batch (batch), name of the distributor (transport_name), name of the receiver (receiver_name), name of the sender (sender_name), test report of the rice (CID), and the stage it is currently in. Upon successful execution, the blockchain network returns a unique tracking number (tracking_number). When the distributor delivers the rice to the factory, the `updateArrivetime()` function is executed to update the logistics status. When the factory receives the rice, it will conduct a comprehensive inspection of the transportation environment, including temperature, humidity, and storage time. In addition, data will be collected by the relevant sensors. The factory then calls the `updateTransRecord()` method, which includes uploading the spectral inspection files to update the transaction record with the latest quality verification data. A similar process applies to a transaction between a distributor and a retail company. The smart contract autonomously enforces quality control during circulation. It processes NIR reports submitted by entities by retrieving and computing them through an Oracle mechanism. A failed verification triggers an automatic update to the batch's blockchain status, which immediately halts further distribution. This automated process ensures that only compliant batches proceed, with complete traceability of all transactions.

3.4. Document processing architecture

IPFS is a decentralized distributed storage protocol, which facilitates efficient and secure data storage for rice supply chain tracing through content addressing, Merkle Directed Acyclic Graph (DAG), and a distributed node network.

Certification documents of all entities and rice inspection files at various stages are stored in IPFS in chunks, with a default chunk size of 256 KB. Each chunk is assigned a CID generated via the SHA-256 cryptographic hash algorithm. Through the Merkle DAG structure, these individual CIDs are integrated into a root CID. Notably, any tampering with a single chunk will irreversibly alter the root CID, thereby enabling effective verification of data integrity.

The root CID is recorded on the blockchain, thereby forming a complementary model combining blockchain endorsement with IPFS storage. In this integrated model, the blockchain is responsible for recording data ownership, transaction history, and the root CID, while IPFS undertakes the storage of original data. This collaborative approach effectively addresses the inherent limitation of blockchain in terms of storage capacity.

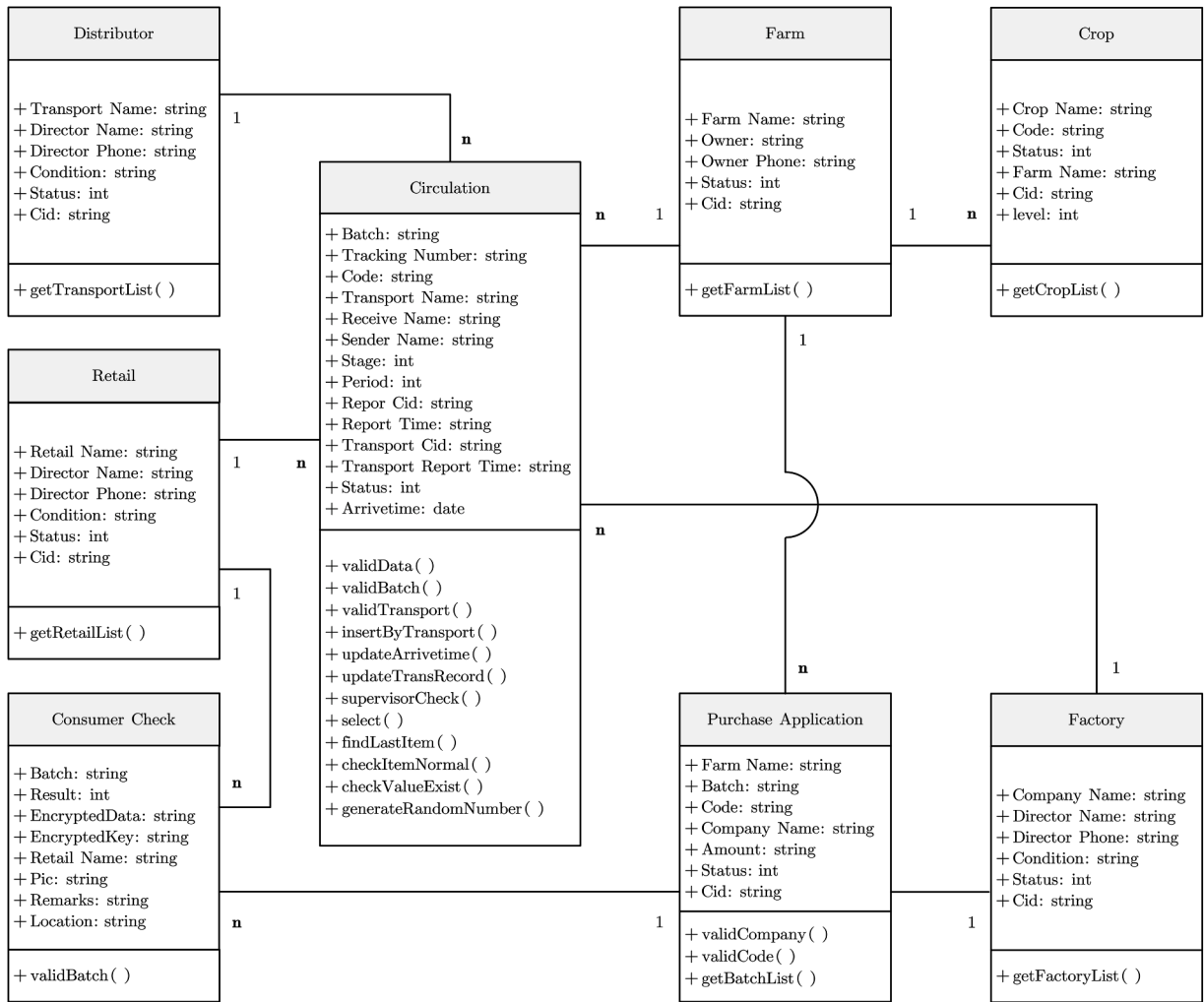


Fig. 3. Blockchain-based underlying data model for rice blending fraud prevention and traceability

3.5. Verification in the process of rice circulation

The proposed model implements quality verification at four critical handover points in the rice supply chain: from farm to distributor, distributor to factory, factory to distributor, and distributor to retailer. This multi-stage verification mechanism is essential for maintaining rice quality, preventing the introduction of substandard or contaminated rice.

3.5.1. Specifications for uploading NIR spectrometer results

The spectral characteristics of rice components like starch, protein, and moisture show unique absorption patterns for NIR light. NIR spectrometers are widely used in agricultural traceability. When NIR light hits rice samples, specific wavelengths are absorbed by these molecules. The reflected light is converted into a wavelength-absorbance curve by a detector. Since absorbance correlates positively with component concentration, this curve can uniquely identify rice quality traits, including purity and origin. Blockchain-based upload specifications of NIR spectrometer results for rice blending fraud prevention are as shown in Fig. 5.

Before model deployment, all participating entities must complete qualification verification and obtain access approval. The regulator then uniformly distributes handheld NIR spectrometers. These devices are pre-installed during manufacturing with asymmetric key pairs, whose public keys are registered centrally with the regulator. The corresponding private keys are securely stored within the device's embedded Hardware Security Module, effectively preventing extraction or duplication [35]. Additionally, each device is assigned a globally unique identifier (spectrometer_id), which maintains a one-to-one correspondence with its public key.

When an operator scans a rice sample, the handheld NIR spectrometer captures the raw spectral signals and associated metadata. This data is then formatted by the embedded gateway, which structures it into a JSON object comprising the spectral matrix and complete metadata, specifically including: operator identity (entity_id), scanning timestamp (scan_time), device ID (spectrometer_id), and spectral characteristic data of rice (spectrum_data). Using its embedded private key, the device generates a digital signature by

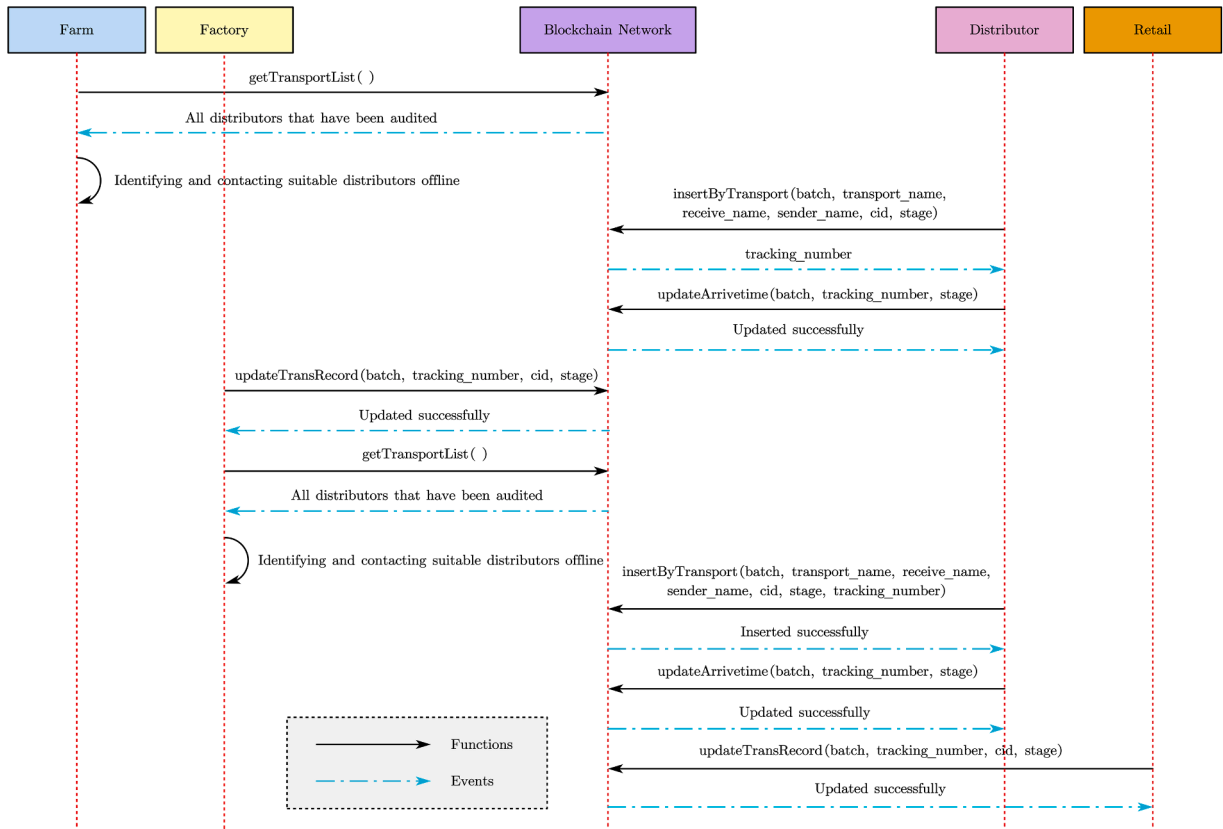


Fig. 4. Blockchain-based sequence flow of rice circulation for blending fraud prevention

encrypting a combined dataset of its identity, a timestamp, and the collected inspection data (signature), and a hash value generated based on the rice spectral data (hash). It is noteworthy that under identical parameter settings, the same device produces strictly consistent “wavelength-absorbance” characteristic curves, a feature ensuring data repeatability under consistent measurement conditions. The device’s digital signature, on the other hand, serves to verify data integrity and non-tamperability, preventing unauthorized alteration or replacement of data during transmission or storage.

Subsequently, smart contracts will retrieve and compare the current scanning data with historical baseline data based on preset logic, enabling dynamic monitoring of rice quality stability by determining whether key quality parameters of rice exceed preset thresholds.

3.5.2. Verification of chainlink oracles

Oracles act as middleware that connects blockchains to the real world, responsible for retrieving and verifying off-chain data. Specifically, by fetching spectral information stored on IPFS, performing comparisons, and submitting the results to the blockchain, they ensure end-to-end data integrity for on-chain applications.

In this model, Chainlink is employed as the decentralized oracle network to facilitate secure off-chain computations. The comparison process between two sets of spectral data begins when a smart contract invokes the oracle’s request function, providing the CIDs of the respective batches. Upon receiving this request, Chainlink nodes utilize the IPFS API to retrieve the corresponding JSON files. These nodes then verify the digital signatures to confirm the files originate from authorized devices before calculating the similarity between the spectral datasets using the cosine similarity algorithm. The computational results, along with the validation status, are relayed back to the smart contract, which executes the appropriate business logic. This approach ensures that spectral data comparisons are accurately performed off-chain, with only the integrity-verified results being permanently recorded on the blockchain. The interaction architecture of NIR spectrometer results, chainlink oracle, and smart contracts is shown in Fig. 6.

3.6. Encryption algorithm designs

To address the issue of data privacy protection on the blockchain, a tiered privacy-protection mechanism has been implemented for on-chain product information. Given the large volume of traceability data and the necessity for public disclosure of most information, the privacy-protection requirements are relatively low. However, in cases involving sensitive customer information, particularly when customers have doubts about their purchased goods, the privacy-protection needs are significantly higher. This design meets

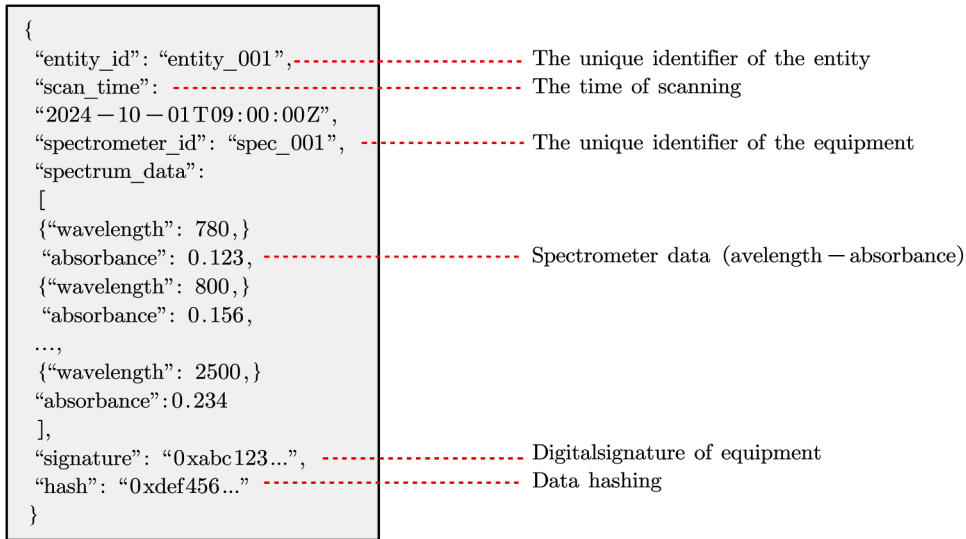


Fig. 5. Blockchain-based upload specifications of NIR spectrometer results for rice blending fraud prevention

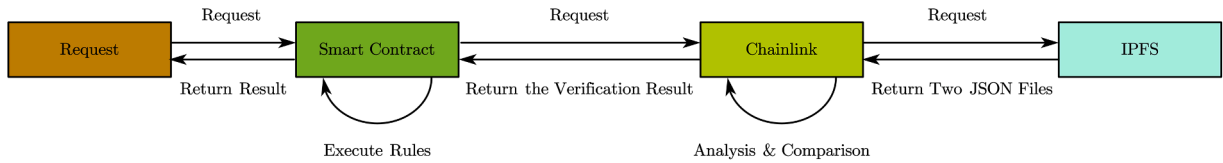


Fig. 6. Interaction architecture of NIR spectrometer results, chainlink oracle and smart contracts

the innovation requirements of our proposed model. Then, the RSA-AES-GCM (Galois/Counter Mode) encryption scheme has been presented for customer information encryption.

Algorithm 1 addresses the encryption of consumer-sensitive information in blockchain models through the integration of RSA and AES-GCM operations. During initialization, a regulatory node generates an RSA key pair. The public key is distributed to consumer nodes; the private key remains exclusively with the regulatory node for decryption, minimizing key management overhead and leakage risks.

When a consumer node requests to upload sensitive data, the algorithm first validates the node's certificate and public key status (Line 1). It then generates an ephemeral AES session key (Line 2), stored temporarily in memory and destroyed after use. The sensitive data is encrypted using AES-GCM (Lines 4–5), producing four output values: username_encrypted, username_auth_tag, phone_encrypted, and phone_auth_tag. These are assembled into an array and serialized for storage efficiency (Lines 7–8).

In the final step, the AES key is encrypted using the consumer's RSA public key, generating the encryptedKey (Line 9). This operation is efficient due to the small key size. The serialized encryptedData and the encryptedKey are then permanently committed to the blockchain, achieving an optimal equilibrium between system performance and data privacy.

Algorithm 2 reverses the encryption procedure of **Algorithm 1**. It first performs a dual verification step (Line 1): it validates that the current node is a regulatory node and checks for the secure availability of the RSA private key.

Upon successful validation, the algorithm proceeds to the key recovery phase. It retrieves the encrypted data package from the blockchain and extracts the encrypted AES key (encryptedKey). This key is then decrypted using the regulatory node's RSA private key (Line 2). To mitigate the risk of exposure, this private key is temporarily stored in memory and never persisted to disk. Subsequently, the encrypted data array is reconstructed through deserialization (Line 4).

With the AES key and encrypted array in place, the algorithm enters the final plaintext recovery phase. It first retrieves all necessary decryption parameters from the array (Lines 5–8). Subsequently, using the AES-GCM algorithm and the recovered AES key, the model step by step decrypts the consumer's username and phone number (Lines 9–10) before returning the result (Line 11).

4. Results and discussions

4.1. Experimental results

User traceability interaction interface design is shown in Fig. 7. User participation significantly enriches the blockchain decision-making model by incorporating diverse data sources, thereby enhancing model trustworthiness, ecological sustainability, and managerial efficiency. In the figure, the orange font indicates that users can download the related report by clicking on it. We have

Algorithm 1 User information encryption.

Input: username, phone
Output: (encryptedData, encryptedKey) or null

```

1: if isConsumer(node) and isExist(public_key) then
2:   aesKey ← generateAesKey();
3:   if aesKey then
4:     (username_encrypted, username_auth_tag) ← encryptInfoByAesGcm(username, aesKey);
5:     (phone_encrypted, phone_auth_tag) ← encryptInfoByAesGcm(phone, aesKey);
6:     if username_encrypted and phone_encrypted then
7:       encryptedDataArray ← [username_encrypted, username_auth_tag, phone_encrypted, phone_auth_tag];
8:       encryptedData ← serialize(encryptedDataArray);
9:       encryptedKey ← encryptKeyByPublicKey(aesKey, public_key);
10:    if encryptedKey then
11:      return encryptedData, encryptedKey;
12:    else
13:      return null;

```

Algorithm 2 User information decryption.

Input: encryptedData, encryptedKey
Output: (username, phone) or null

```

1: if isRegulators(node) and isExist(private_key) then
2:   aesKey ← decryptKeyByPrivateKey(encryptedKey, private_key);
3:   if aesKey then
4:     encryptedDataArray ← deserialize(encryptedData);
5:     username_encrypted ← encryptedDataArray[0];
6:     username_auth_tag ← encryptedDataArray[1];
7:     phone_encrypted ← encryptedDataArray[2];
8:     phone_auth_tag ← encryptedDataArray[3];
9:     username ← decryptInfoByAesGcm(username_encrypted, aesKey, username_auth_tag);
10:    phone ← decryptInfoByAesGcm(phone_encrypted, aesKey, phone_auth_tag);
11:    return username, phone;
12:  else
13:    return null;

```

prototyped this solution based on FISCO BCOS. The experiment runs on an Intel(R) Xeon(R) Platinum CPU 2.5GHz with 4 cores and 8GB RAM. We note that other federated blockchain platforms can also be used.

Data throughput, measured in Transactions Per Second (TPS), refers to the number of transactions the blockchain can process each second, and is an important measure of the efficiency and performance of a traceability model. We tested the storage and querying of rice product information by different participants in the blockchain network. They can be roughly categorized into traceability query, transaction query, traceability storage, and transaction storage. In order to ensure the reliability of the experimental data, we conducted 100 experiments and took the average value. Throughput test results of the blockchain system in multiple scenarios are shown in Fig. 8.

The experiments confirm a linear growth in TPS with increasing data volume, demonstrating the model's robust scalability and stability. (The detailed test results can be found in [36].) Note that our experiment measures the average throughput of all actors operating within a single entity node in a blockchain network, and data throughput will be significantly higher when multiple entities are working in their respective nodes.

4.2. Comparative experiments

To demonstrate the effectiveness of our results, a comparative experiment was set up under identical configuration and concurrency conditions. A lightweight script was implemented using Golang and MySQL to perform "save" operations and "select" operations on database tables. The queries were optimized to utilize database indexes, ensuring maximum efficiency.

Fig. 9 presents the throughput test results of the traditional database control group. While the highly optimized MySQL control group achieves marginally higher peak throughput in specific instances, our blockchain model attains a comparable performance level despite the inherent overhead of distributed consensus and cryptographic verification.

The performance curve of the control group exhibits significant fluctuations, revealing its limitations in handling lock contention under concurrent conditions. In contrast, our model not only maintains comparable peak throughput but also delivers stable performance through its deterministic transaction workflow.

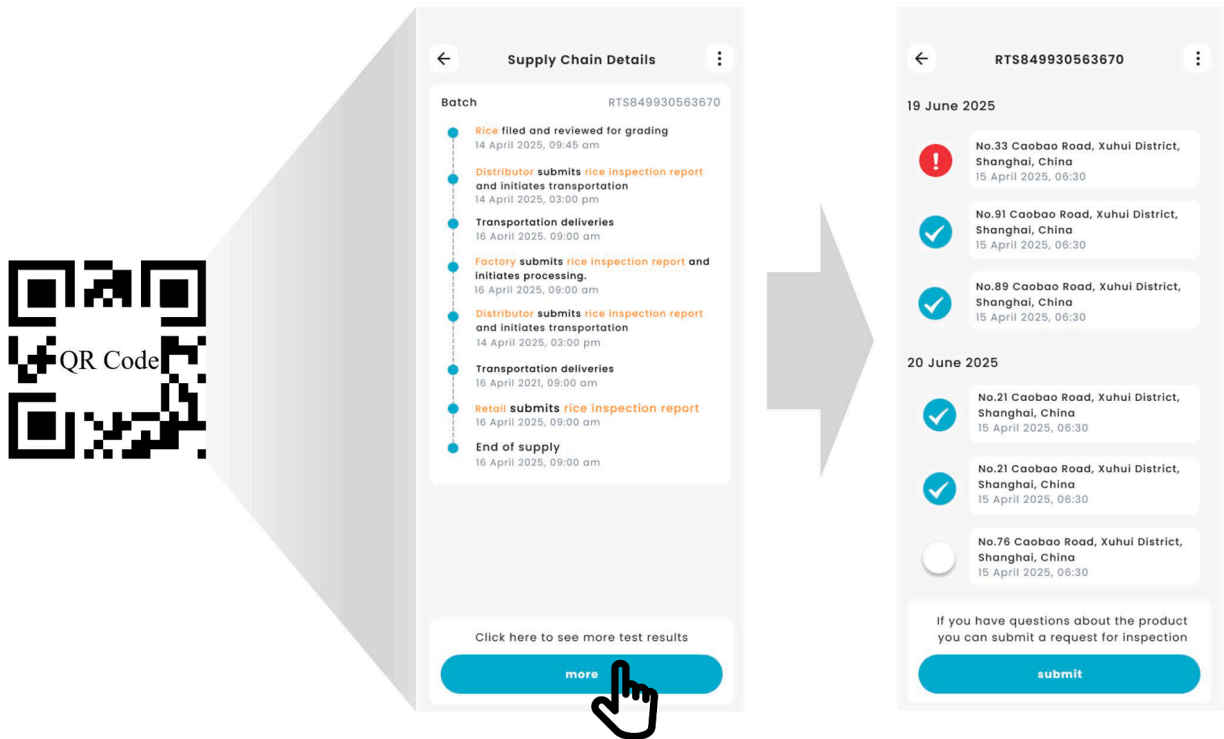


Fig. 7. User traceability interaction interface design

Table 1
Comparison of rice blending fraud prevention traceability model with existing models

	Our scheme	Zhang et al. [24]	Salah et al. [37]	Ren et al. [38]	Guan et al. [39]
Storage Design	✓	✓	✓	✓	✓
Quality Inspection	✓	✓	✓	✗	✓
Privacy Protection	✓	✗	✗	✗	✓
Data Sharing	✓	✓	✓	✓	✓
User Decision-Making	✓	✓	✗	✗	✗

Our model achieves performance comparable to conventional solutions while managing the full complexity of a distributed ledger. This performance parity under complex conditions validates the effectiveness of its design and implementation.

4.3. Comparison with the existing models

Comparison of rice blending fraud prevention traceability model with existing models is shown in Table 1. Our proposed traceability model for preventing rice adulteration has been systematically evaluated against existing models across multiple critical dimensions, including storage design, quality inspection, privacy protection, data sharing, and consumer participation in decision-making. The results demonstrate the outstanding performance of our model across the comprehensive set of evaluation metrics.

It should be noted that while most existing studies focus on quality inspection, they are primarily concentrated on the initial supply chain stages (e.g., farm-level planting records or in-transit sensor data) and largely neglect subsequent phases. In contrast, our scheme extends quality inspection to cover potential points of adulteration across the entire supply chain. This comprehensive approach advances research on full-chain quality monitoring and achieves more thorough rice quality control.

4.4. Model extensibility

4.4.1. Scalability enhancement through Layer-2 architecture

In current blockchain applications, layered architecture has been widely adopted to enhance model performance. A typical two-layer structure isolates testing and production environments to balance functional validation and commercial deployment. A common functional layering strategy allocates high-security, low-frequency operations to the main chain (Layer-1), while the sidechain (Layer-2) is dedicated to high-throughput tasks. Data integrity is maintained as the sidechain periodically submits data hashes to Layer-1.

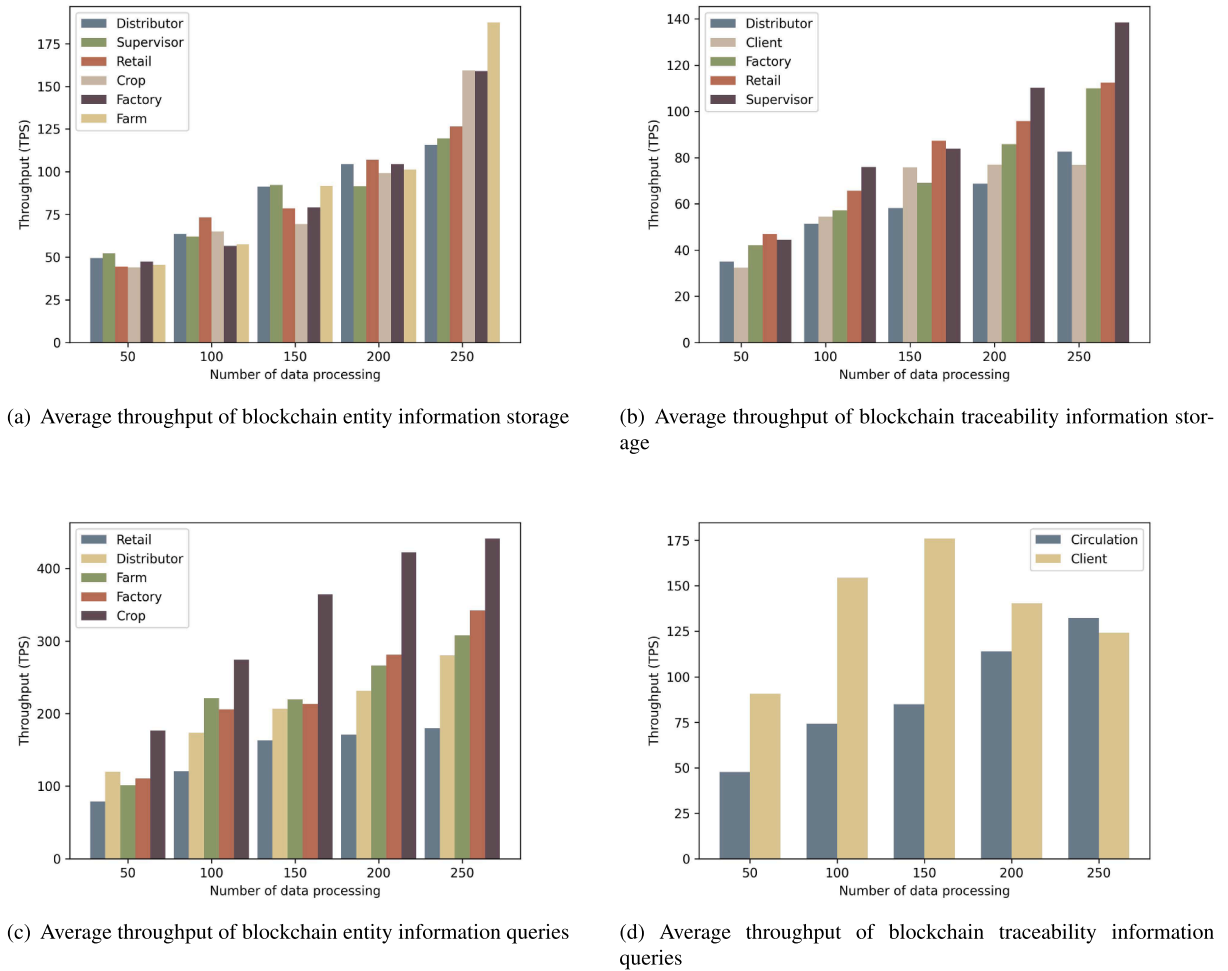


Fig. 8. Throughput test results of blockchain system in multiple scenarios

Based on this architecture, the proposed rice traceability model also adopts a layered design: Layer-1 serves as the core trust layer, managing participant identities, rice ownership status, and dispute arbitration; Layer-2 acts as the business processing layer, performing real-time quality inspection, distributed data storage, and efficient query responses. The specific workflow is as follows: after a farmer submits a rice registration request on Layer-1, a regulator stores detailed inspection data via Layer-2 into IPFS, obtains a CID1, and binds CID1 to the corresponding rice asset on Layer-1. During circulation, when a distributor receives the rice, an ownership transfer process is triggered on Layer-1, while quality verification is performed via Layer-2. Real-time inspection data is stored in IPFS to obtain CID2, and a Chainlink oracle is used to retrieve both the baseline CID1 and real-time CID2 from IPFS for comparison. The verification result is returned to the Layer-1 smart contract, which determines whether to proceed with the ownership transfer. Consumers can efficiently retrieve full traceability information via Layer-2. Should a quality dispute arise, it is recorded on the Layer-1 smart contract to trigger regulatory intervention, and the final resolution is updated to Layer-1 upon conclusion.

In this model, Layer-2 implementation is not framework-dependent. Given the high performance of FISCO BCOS, WeCross, or a custom aggregation layer may be prioritized, focusing on batch data processing and hash anchoring.

4.4.2. Achieving scalability via sharding

In the current development of blockchain technology, sharding has emerged as a critical scaling solution, whose core principle lies in distributing model state and computational tasks across multiple independent shards for parallel processing. Prominent blockchain projects such as Ethereum 2.0 and Zilliqa have adopted this design philosophy. On the FISCO BCOS platform, sharding can be implemented through its multi-group architecture, where each group operates as an independent chain maintaining its own state and consensus mechanism. Cross-group interactions are treated as inter-shard communication, requiring support from cross-chain contracts or middleware.

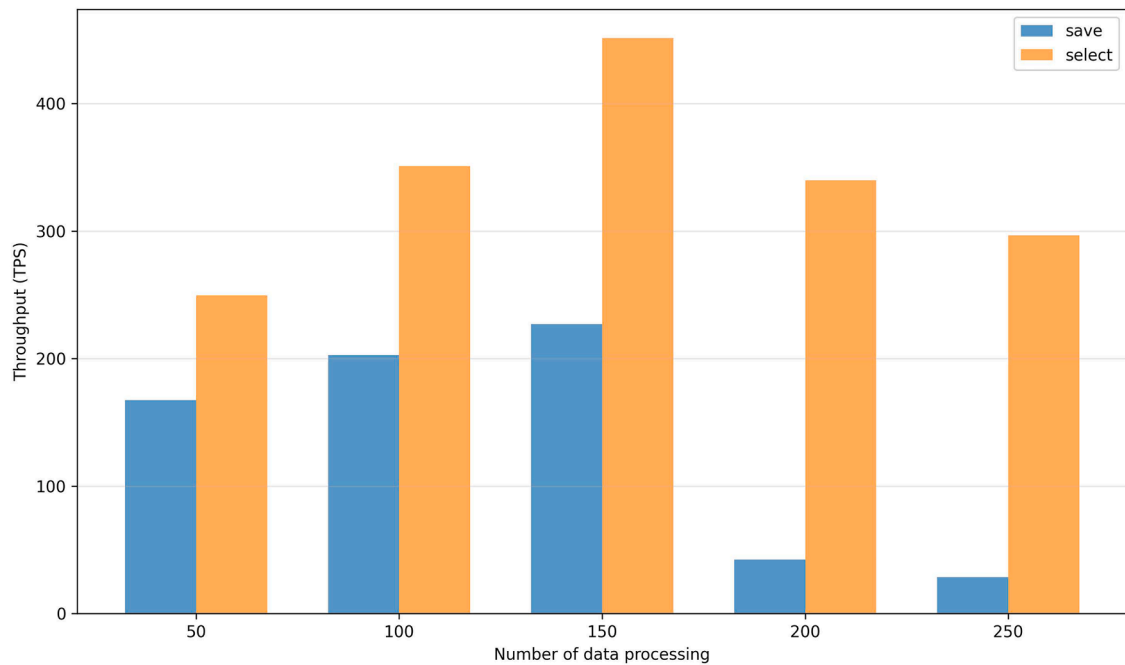


Fig. 9. Throughput test results of the traditional database control group

The proposed traceability model is structured around four functional shards: Management, Production, Processing, and Circulation. The Management Shard maintains inter-shard relational links for each batch, enabling unified cross-shard traceability queries, in addition to its core functions of entity registration and dispute resolution. The Production Shard notarizes planting information and initial quality data on the blockchain. The Processing Shard oversees operations such as purchase requests and quality analysis. During distribution, the Circulation Shard records ownership transfers and sensor-based transport reports, tracking the real-time status of batches as they move through the supply chain.

To reduce model complexity, the model adopts a simplified cross-shard interaction approach. Instead of directly transferring assets across shards, batch information is independently recorded within each shard, with only the global transfer path maintained by the management shard. While this method increases integration overhead during the query phase, it effectively circumvents the technical challenges associated with cross-shard asset transfer, thereby simplifying model implementation.

5. Conclusion

This paper presents an innovative blockchain-based traceability framework designed to resolve the critical challenge of blended-quality rice circulation in supply chains. The proposed system integrates two fundamental technological mechanisms: dedicated NIR spectrometers deployed at strategic transfer nodes for objective quality verification, coupled with a consumer participation framework for enhanced oversight, effectively overcoming the industry's persistent limitation of visual quality assessment. Simultaneously, a carefully engineered hybrid encryption scheme balances robust cryptographic protection with operational efficiency, ensuring both data security and system performance. The architecture demonstrates significant extensibility to various products requiring non-visual quality authentication, with comprehensive stress testing across four key operational dimensions (i.e., traceability queries, transaction processing, traceability storage, and transaction storage), confirming the framework's practical utility, reliability, and scalability for real-world deployment scenarios.

CRedit authorship contribution statement

Tao Peng: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Langfeng Wang:** Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Data availability

I have shared the link to my data in reference section.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used DeepSeek and ChatGPT to improve language and readability. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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